

GeoStory: Usability Testing and Evaluation

Lab 7 Test Report

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Abstract

We tested our Lab 6 build of GeoStory in three ways: user testing with five people from outside our team, a heuristic evaluation against Nielsen's 10 heuristics, and an accessibility check. All five users completed both key tasks, and the average SUS score was 79 out of 100. The main problem was not in the tasks themselves but in understanding the website: users did not know what GeoStory is or how its five *threads* connect stories across time. This report explains the test plan, the findings and our plan for Lab 8.

1. Test Plan

1.1 Tasks

We used the two key user tasks from Lab 5:

- **T1 – Navigate time to discover activity:** “Go to 1300 CE and discover the story in West Africa.” (Success: the Mansa Musa story card is open.)
- **T2 – Explore a chronicle:** “Read the full story after discovering that chronicle.” (Success: the full chronicle page is open.)

These two tasks cover the core of GeoStory: the timeline, the map pins, the story card and the full chronicle.

1.2 Users

Five students from outside our team (U1–U5). None had seen GeoStory before. Five users is the minimum asked for, and it is also the number usually suggested for finding most usability problems in a small test.

1.3 Procedure

1. The laptop was given to the user with GeoStory open at the first timeline stop (Big Bang). We did *not* show the intro screen, because it has a “Jump to 1300” shortcut that would have given away Task 1.
2. We read out T1, started a stopwatch, and watched without helping. We noted where the user hesitated or got stuck. We gave a hint only if the user was stuck for a long time.
3. We repeated this for T2.
4. The user then explored the website freely and told us what they understood about it and what they would change.
5. Finally, the user filled in the 10-question SUS form (Google Forms) with one open question for suggestions. Separately, both team members did the heuristic evaluation on their own and then merged their lists, and we did the accessibility check together.

1.4 Metrics and why we chose them

- **Completion** (Yes / With help / No): shows whether the task can be done at all.
- **Time on task:** shows how easy the task is. A long time means the user is searching or confused.

- **Confusion points:** where users paused or needed help. These tell us *what* to fix.
- **SUS score:** a standard, quick questionnaire that gives one overall usability score, which we can compare after Lab 8.

2. Findings

2.1 User testing results

Task 1 took much longer than Task 2 and varied a lot (13.5 s to over 3 min). What we saw:

- U1 clicked “1300” on the timeline directly, but took some time to notice the pin and to understand that it has to be clicked.
- U2 moved era by era with the arrows and did not know at first that the timeline stops can be clicked directly.
- U5 could not find the story alone and needed a hint.

Task 2 was fast for everyone (avg 8.4 s), because the “Read the full chronicle” button is clearly visible on the story card.

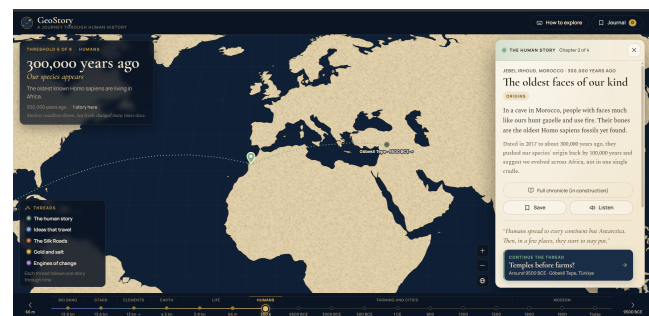


Figure 1: Story card open (Task 1 result). The era card (top-left) and the threads panel (bottom-left) stay open and crowd the map (H8). The threads panel shows only names, with no chapters (U1). The timeline years at the bottom are small (U2), and the linked pin at the left edge is cut off.

What users said during free exploration. The biggest problem was understanding the website as a whole:

- Users did not understand that there are **5 threads**, that each thread links several stories across eras, and that a thread can be followed by matching its colour or by using the next/previous chapter button on a story card.
- Users did not know what GeoStory is or how to use it, because there is no “what is this” or “how to use” section at the start. (This was partly caused by our choice to skip the intro screen, but the intro screen also does not explain threads.)
- The **years on the timeline are too small** to read.

Suggestions from the SUS form: show pop-ups for next chapters; bigger icons; clearer zoom in/out icons (the +/– buttons exist but were not noticed); more stories and pictures; maps that match the time period more accurately; nar-

Table 1: Heuristic, accessibility and user-testing issues. Severity: 0 = not a problem, 1 = cosmetic, 2 = minor, 3 = major, 4 = catastrophe. “UT” means the issue was also seen in user testing.

#	Screen / Step	Problem	Heuristic / Check	Sev.
U1	Threads panel, story card	Users did not understand the 5 threads, that each has chapters across eras, or how to follow one (UT: most users)	6. Recognition rather than recall; 10. Help	3
U2	First screen	No “what is GeoStory / how to use it” section; users did not know the timeline stops can be clicked (UT: U2, U5)	10. Help and documentation	3
U3	Timeline	Years on the timeline are too small to read (UT)	8. Aesthetic; Accessibility (text size)	2
H1	Map, while moving in time	Pins are small; users may not notice new pins or that they must be clicked (UT: U1)	1. Visibility of system status	2
H2	Timeline labels	“66 m”, “300 k”, “13.8 bn” are unclear (“m” can read as metres)	2. Match with real world	2
H3	Journal	A saved story removed by mistake cannot be brought back	3. User control and freedom	2
H4	Journal	The same × icon closes the panel and removes a saved story	4. Consistency and standards	2
H5	Journal	A saved story is removed instantly, with no check	5. Error prevention	2
H6	Map controls	The globe icon (reset view) is unclear; +/– zoom buttons were not noticed (UT)	6. Recognition rather than recall	2
H7	Finding a story	No search; users must travel the timeline to find a chronicle	7. Flexibility and efficiency	2
H8	Story card open	Era card and threads panel stay open and crowd the map; linked pin can go off-screen (Fig. 1)	8. Aesthetic and minimalist design	3
H9	–	No problems found; error messages are in plain language	9. Recover from errors	0
A1	Whole app at 200% zoom	Zoom is inconsistent: header, era card and timeline enlarge, map redraws smaller, threads panel disappears	Accessibility: text size	3
A2	Whole app	No problems found; Lighthouse accessibility score 100 (Fig. 2)	Accessibility: contrast	0
A3	Both tasks	Both tasks can be done with the keyboard only	Accessibility: keyboard	0
A4	Buttons, controls	No problems found (all buttons have labels)	Accessibility: labels	0

Table 2: Task completion and time (s = seconds).

User	T1	Time	T2	Time
U1	Yes	13.5 s	Yes	2 s
U2	Yes	48 s	Yes	6 s
U3	Yes	2 min 30 s	Yes	5 s
U4	Yes	1 min 9 s	Yes	7 s
U5	With help	3 min 11 s	Yes	22 s
Result	4/5 alone (5/5 total)	avg 94 s median 69 s	5/5	avg 8.4 s median 6 s

ration in more languages; more Indian history. U1 suggested showing the number of chapters beside each thread and listing the chapters on hover.

2.2 SUS questionnaire

SUS scores are shown in Table 3. The standard score has an average of **79.0**, which is above the usual benchmark of 68.

Table 3: SUS score per user (0–100).

	U1	U2	U3	U4	U5	Avg
Standard SUS	87.5	82.5	77.5	77.5	70.0	79.0
Without Q6	86.1	83.3	77.8	86.1	77.8	82.2

Limits of this data. We treat the SUS result as a rough signal only:

- Some users answered quickly with 2s and 4s without much thought.
- Two users gave a 5 to Q6 (“too much inconsistency”), which clashes with their other answers. They told us they read it as “consistency”. So we also show the score without Q6 (the other 9 answers scaled to 100).
- Five users is a small sample.

The task times and what users said are more reliable evidence than the SUS numbers.

2.3 Heuristic evaluation and accessibility

Table 1 lists all issues from the heuristic evaluation (H), the accessibility check (A) and user testing (U). User testing confirmed several heuristic issues. For example, the small pins (H1) slowed down U1, and the unnoticed zoom buttons (H6) came up in the SUS suggestions.

For accessibility, Chrome Lighthouse gave a score of 100 (Fig. 2), and keyboard-only use worked for both tasks. Lighthouse cannot test everything, though. Our manual 200% zoom test found a real problem (A1): content is lost because the threads panel disappears.

3. Improvement Plan

The main lesson is that users could *do* the tasks but did not understand *what GeoStory offers*, especially the threads. The fixes are ranked by severity:

Severity 3 – fix in Lab 8

1. **New HELP page as the intro screen (U1, U2).** Re-design “How to explore” into a *Help* page. It explains

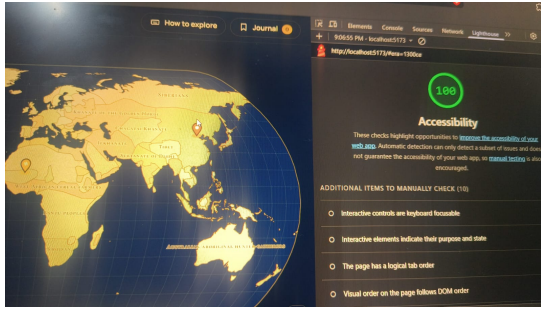


Figure 2: Chrome Lighthouse accessibility audit at the 1300 CE stop: score 100.

what GeoStory is, what the timeline, pins, story cards, chronicles and journal do, and what the 5 threads are and how to follow them. It uses an annotated screenshot with arrows and ends with an **Enter** button. The page can be opened again at any time from the header.

2. **Better threads panel (U1).** Show the number of chapters beside each thread (e.g. “Gold and salt · 4”). On hover, list the thread’s chapters with their years, and let the user click one to go straight to it.
3. **Less clutter when a story is open (H8).** Shrink the era card and threads panel, and keep linked pins inside the view.
4. **Fix 200% zoom (A1).** Turn the threads panel into a button instead of hiding it, and make map labels and pins scale with zoom.

Severity 2 – fix in Lab 8 if time allows

5. Larger timeline years and full labels such as “66 million years ago” on hover (U3, H2).
6. Bigger, more visible pins (H1).
7. Tooltips (“Zoom in”, “Reset view”) and clearer map controls (H6).
8. Journal: a separate trash icon for removing, plus an “Undo” message (H3, H4, H5).
9. Search box to jump directly to a story (H7).

Later labs (content requests). More stories and pictures, Indian history, narration in more languages, and checking that map borders match each period. These are content requests, not usability problems, so they belong to the content work in later labs.